A Study of Model Based and Model Free Offline Reinforcement Learning

Indu Shukla
Information Technology Laboratory
U.S. Army Engineer Research and Development Center
3909 Halls Ferry Rd, Vicksburg MS, USA
Indu.Shukla@usace.army.mil

Haley R. Dozier
Information Technology Laboratory
U.S. Army Engineer Research and Development Center
3909 Halls Ferry Rd, Vicksburg MS, USA
Haley.R.Dozier@usace.army.mil

Althea C. Henslee
Information Technology Laboratory
U.S. Army Engineer Research and Development Center
3909 Halls Ferry Rd, Vicksburg MS, USA
Henslee.C.Althea@usace.army.mil

***Abstract*— Reinforcement learning (RL) has received considerable attention for building autonomous systems with trained agents that interact with environments to learn optimal behavior. The RL requires a fundamentally online learning paradigm, one of the biggest obstacles for the widespread adoption of RL in scaling to many real-world scenarios. In this study we have explored the method of data collection and have applied offline model-free and model-based RL methods on classical OpenAIGym Cart-Pole environment. In this approach offline algorithms find a good policy from previously collected dataset.**

Keywords—Offline Reinforcement Learning, Model-Free methods, Model-Based methods, OpenAI Gym, data collection

I. Introduction

Reinforcement Learning (RL) is a form of machine learning where an agent learns the optimal action for a given task through its repeated interaction with a dynamic environment. RL generated renewed interest among researchers who are now successfully applying this model to solve very complex problems which were considered intractable earlier [1].

RL can also be divided between model-based methods and model-free methods. Model-based methods use both the transition function and the reward function to learn the transition dynamics of the environment and predict the dynamics of the environment. Models are used in planning by deciding the course of action for the consideration of possible future situations. Model-free methods only train a reward-maximizing policy on the environment without estimating the dynamics of the environment. Model-free methods do not use either a transition function or a reward function, learning instead from experience by estimating the value function or policy, and thus are trial and error learners [2]. However, the fact that RL algorithms provide a fundamentally online learning paradigm is also one of the biggest obstacles to their widespread adoption. In many settings, online interaction is impractical because data collection is expensive and dangerous. Success of machine learning methods across the range of practically relevant problems has attributed to data-driven RL methods, which utilize previously collected offline data [3]. Offline RL methods have received less attention in the literature than online RL methods, and present a more challenging setting. In this paper we have applied offline experimental model-based and model-free method for controlling a Cart-Pole system.

Our goal is to apply offline RL in one- and two-dimensional mission problem in continuous action space. Agents trained by offline RL methods strictly rely on static datasets without the possibility of exploration. But we believe that by using simulation software, it would be possible to discover new transitions and retrain the agent with new generated data. Training the agent for policies with offline, data-driven RL in simulation and then transferring these policies into the real world would increase capability, automation, and integration and therefore reduce the labor-intensive manual planning.

II. Methods

A. Cart-Pole System

For our experiment, we have used a continuous version of the Cart-Pole environment written in OpenAI Gym style [3]. It consists of a cart and a vertical bar attached to the cart using a passive pivot joint. The objective is to prevent the vertical bar from falling by moving the cart left or right. The dynamical system of the continuous version of Cart-Pole is very similar to that of the discrete version of Cart-Pole, the only difference being valid actions are any number in the range [-1,1], applied as a multiplicative factor to the total force. The agents receive the reward of +1 for every timestep that the pole remains upright [13].

B. Offline Reinforcement Learning

Offline RL is a re-emerging area of study that aims to learn policies entirely from a large batch of previously collected data without the dynamic environmental interaction [5]. This approach has potential to make tremendous progress in a number of real-world decision-making problems where active data collection is expensive. In offline RL, the agent must learn the best policy from the provided static dataset and construct a policy that attains the largest possible cumulative award. However, offline RL is still a challenging approach because the agent no longer has the ability to interact with the environment to collect additional transitions and improve exploration [3,7].

C. Model-Free and Model-Based Offline Learning

There are both model-based and model-free offline RL methods [14]. In the current work, we have used the model-based Offline Policy Optimization (MOPO) method, which utilizes conservative value estimates to provide analytic bound on performance. In this approach, a penalty is given for visiting states supported by the incorrect model [6].

Model-free offline methods constitute the majority of algorithms in RL literature. In this paper we have used offline Conservative Q-Learning (CQL) approach to train a Qfunction with a separate policy which maximizes the expected Q-value [8]. In this paper, we used d3rlpy, an open-sourced offline deep RL library for Python [7].

III. Results and discussion

We evaluated the model-based and model-free offline learning approach on OpenAI [13] gym style Cart-Pole game system. We performed experiments and evaluated results on existing offline approaches.

d3rlpy supports a number of offline model-based and model-free deep RL algorithms as well as online algorithms. For this study we have used both model-based and model-free techniques for balancing the Cart-Pole system. We have used the Stable-Baseline SB3 utility helper to create the dataset [9]. We have also used d3rlpy MDPDataset module for converting to the offline RL dataset, which works with d3rlpy models.

The MDP dataset consists of a list of object episodes. The hierarchy of the Episode module and Transition module is convenient for episode-wise data division for training and testing. We have used the continuous version of the Cart-Pole environment written in an OpenAI Gym style environment, from the MBRL pets implementation [10].

Our goal is to determine if a generated offline dataset can be used for RL to balance the Cart-Pole system using model-free and model-based algorithms. We then compare the performance of model-free and model-based models from the generated datasetset for 6000 timesteps. We evaulate on an average reward accumulated over x number of episodes. We test the trained agent with access to the true environment to measure the performance of offline RL.

To evaluate the offline model-free agent, we have used CQL, which regularizes the Q-values during training. Figure 2 shows that the model-free algorithm trained for 100 epochs is able to solve the problem in 100 episodes, reaching the max reward of 400. For 500 episodes, the agent performed even better, exceeding more than 400 reward for a few cases.

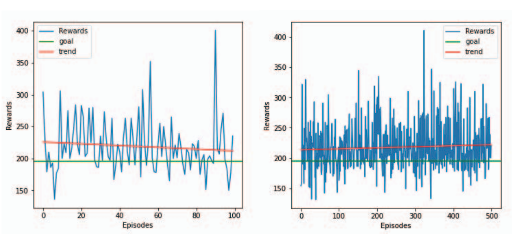

Figure 1. Performance of model free algorithm.

We found that the training using an offline model-based approach is slower than training using a model-free algorithm. This difference could be because the Cart-Pole system is simple and might not require complex architecture to train the agent in simple game environments. We have considered training dynamic models for 10 epochs and model-based models for one epoch. Even with a small training set, the model-based algorithm scaled well, reaching a reward of 550 for 500 episodes as shown in Figure 2.

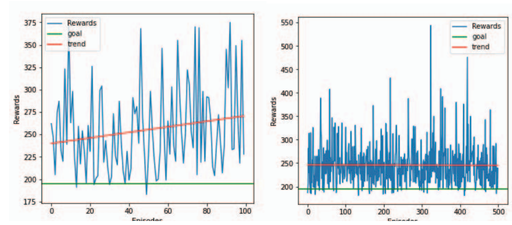

Figure 2. Model based algorithm reward

IV. Conclusions and future outlook

In the experiments, both offline model-based and model-free showed promising and competitive results. In our experiments, the offline model-free method outperforms the model-based method, implying the usefulness of offline learning for more complex scenarios.

Model-based offline approaches extend the use of RL agents. Therefore, increasing the depth and breadth of operational analysis will provide new insights on current challenges in military missions. The key research question that will be addressed in our future work under the mission planning will be an implementation of offline model-based approach. We believe that using this approach the agent will effectively exhibit intelligent mission planning behaviors without requiring training data on billions of possible combinations of situations.

References

[1] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, “Deep reinforcement learning: A brief survey,” IEEE Signal Processing Magazine, vol. 34, pp. 26–38, 2017.

[2] R.S. Sutton and A.G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, Cambridge, MA 2018. http://incompleteideas.net/book/bookdraft2017nov5.pdf

[3] S. Levine, A. Kumar, G. Tucker and J. Fu. Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems. ArXiv abs/2005.01643 (2020).

[4] P. Swazinna, S. Udluft, D. Hein, T. Runkler, “Comparing model-free and model-based algorithms for offline reinforcement learning,” IFAC-PapersOnLine, vol. 55, pp. 19-26, 2022.

[5] T. Yu, G. Thomas, L. Yu, S. Ermon, J. Zou, S. Levine, C. Finn, and T. Ma, Mopo: Model-based offline policy optimization. 34th Conference on Neural information proceeding Systems (NeurIPS 2020), Vancouver, Canada.

[6] T. Seno, M. Imai, D3rlpy: An offline deep reinforcement learning library. ArXiv abs/2111.03788 (2021).

[7] A. Kumar, A. Zhou, G. Tucker, S. Levine, “Conservative q-learning for offline reinforcement learning,” 34th Conference on Neural information proceeding Systems (NeurIPS 2020), Vancouver, Canada.

[8] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann, “Stable-baselines3: Reliable reinforcement learning implementations,” J. Mach. Learn. Res. Vol. 22, pp. 1-8, 2021.

[9] P. Luis, B. Amos, A. Zhang, N. Lambert and R. Calandra, “MBRL-Lib: A Modular Library for Model-based Reinforcement Learning.” *ArXiv* abs/2104.10159 (2021).

[10] C. Yildiz, M. Heinonen, H. Lahdesmaki, “Continuous-time model-based reinforcement learning,” Proc. Mach. Learn. Res. Vol. 139, pp. 12009-12018, 2021.

[11] OpenAI Gym. Toolkit for developing and comparing reinforcement learning algorithms. https://gym.openai.com/

[12] C. Gulcehre, Z. Wang, A. Novikov, T.L. Paine, S.G. Colmenarejo, K. Zolna, R. Agarwal, J. Merel, D. Mankowitz, C. Paduraru, G. Dulac-Arnold, J. Li, M. Norouzi, M. Hoffman, N. Heess, N. de Freitas, “RL unplugged: A suite of benchmarks for offline reinforcement learning,” 34th Conference on Neural information proceeding Systems (NeurIPS 2020), Vancouver, Canada.